ISSUE 001 · THE CURIOSITY EDITION

# The Guided Planet

THE NEWSPAPER FOR READERS 7 TO 12 AND THEIR GROWN UPS

FREE!

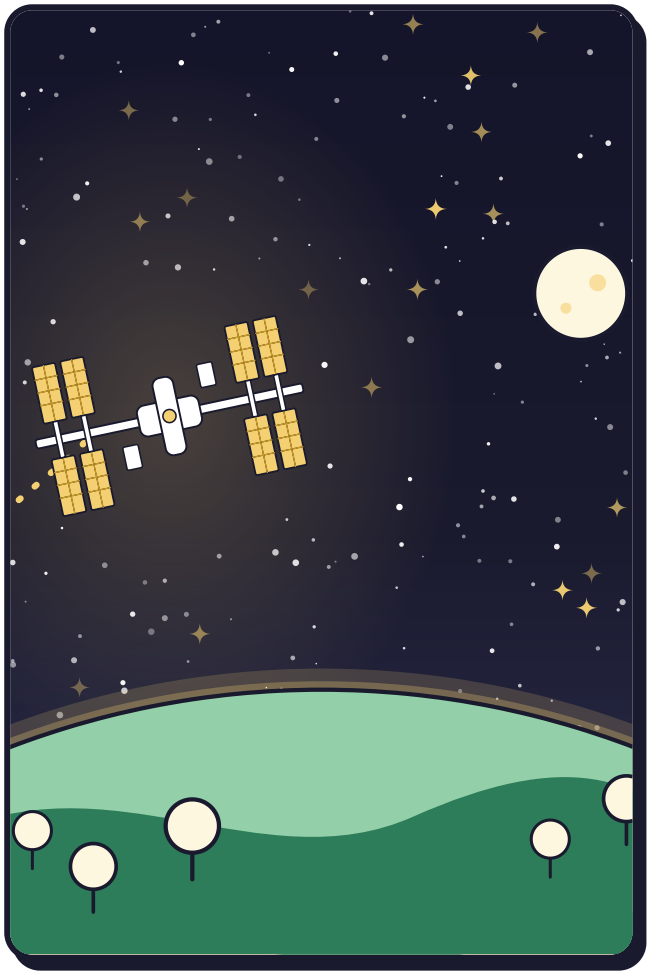
NOVEMBER 2026

**A little more wonder. A lot more understanding.**

FRONT PAGE NEWS

## The good side of screens.

This month: children's code that runs in space, young inventors fixing real problems, and a paper game where you get to be the algorithm.



INSIDE


3 The children's code travelling above Earth

4 Young inventors, real problems

6 Not all screen time is the same

9 Become the Algorithm


MEET THE PLANET FRIENDS



FROM THE EDITOR'S DESK

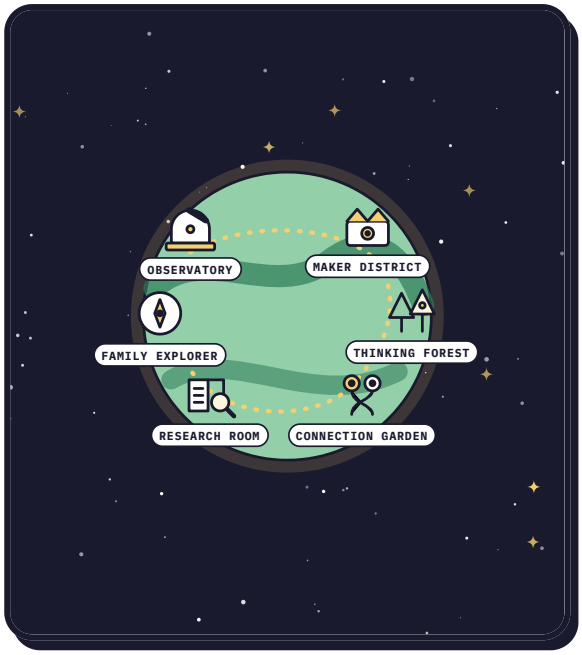
# Welcome to The Guided Planet

Hello, and welcome to our very first issue. This is a newspaper about the good side of screens: the brilliant things children make, discover and share with technology, and the clever questions worth asking along the way.

Every month we visit six places on our planet. The Discovery Observatory is for space and science, and the Maker District is for inventing. In the Thinking Forest we work out how technology really works. The Connection Garden is all about friendship, the Research Room checks what scientists actually know, and the Family Explorer is full of things to do together.

Every fact has its source printed on the same page, so you can always check where it came from. Checking is a superpower. We hope you use it.

**Justin Phillips**
EDITOR IN CHIEF, THE GUIDED PLANET




YOUR MAP: SIX PLACES TO EXPLORE, EVERY ISSUE

HOW TO USE THIS PAPER

- A number in a little circle matches a source in “Where this came from” at the foot of the page.
- You need a pencil, some colours and scissors. No phone and no account.
- Collect a stamp on page 12 for every activity you try.

INSIDE THIS ISSUE

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4 Young inventors, big ideas
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6 Not all screen time tells the same story
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BLOOP SAYS
Hi, I'm Bloop! I love building things, although I am nearly always one piece short. I'll pop up all through the paper with questions. Ready?



The Discovery Observatory · SPACE NEWS

# The children's code travelling above Earth

Young people's code runs on the International Space Station. Last season, 25,707 of them were due to have a program run in space.[4] This season, you could be one of them.

On board the International Space Station are small computers called Astro Pis, and young people on Earth get to program them.[1]

In Mission Zero, run by the European Space Agency with the Raspberry Pi Foundation, you write a short Python program that shows a picture on an Astro Pi, using its colour sensor to pick the background.

It all happens in a web browser.[1]

Young people aged 19 or under at a UK school or club can enter, alone or in a team of up to four, with a mentor such as a teacher or parent.[3] Every team whose code runs gets a certificate showing where the space station was at the time.[1]

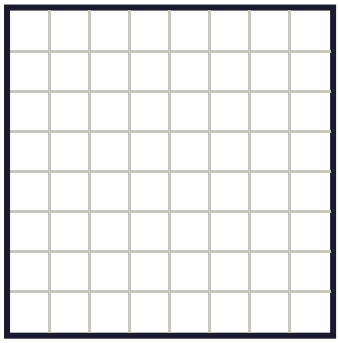
FACT FILE

**22 March 2027**
The deadline for Mission Zero entries, at 12 noon CET.[2]

**9 to 16**
The ages Mission Zero is ideal for, though you can enter up to 19.[5][3]

**24,695**
Young people who took part in Mission Zero last season, 2025 to 2026.[4]

ACTIVITY · NO SCREEN NEEDED



64 SQUARES · 8 ACROSS · 8 DOWN

## Design your space picture

The Sense HAT, the board made for the space station's Astro Pi computers, has a screen of 64 coloured lights in an 8 by 8 grid.[6] Plan your picture here first.

1. Pick up to four colours from the key, or choose your own.
2. Colour one square at a time. Each square is one light.
3. Keep it bold: a planet, a pet, a smiley face or your initial.
4. With a grown up, use the free Mission Zero guide to turn it into code. First names only, the rules say.[1][2]

BUTTER · LEAF · SKY · NIGHT

WHERE THIS CAME FROM


1. **ESA Education and the Raspberry Pi Foundation**, Mission Zero. astro-pi.org
2. **ESA Education and the Raspberry Pi Foundation**, Mission Zero guidelines. astro-pi.org
3. **ESA Education and the Raspberry Pi Foundation**, Mission Zero eligibility. astro-pi.org
4. **Raspberry Pi Foundation**, An astronomical anniversary: young people's code heads to the International Space Station, 22 April 2026. raspberrypi.org
5. **Raspberry Pi Foundation**, Launching Astro Pi 2026/27: code your way to the International Space Station, 14 September 2026. raspberrypi.org
6. **Raspberry Pi**, Sense HAT. raspberrypi.com

The Maker District

YOUNG INVENTORS

# Young inventors. Big ideas. A world of possibilities.

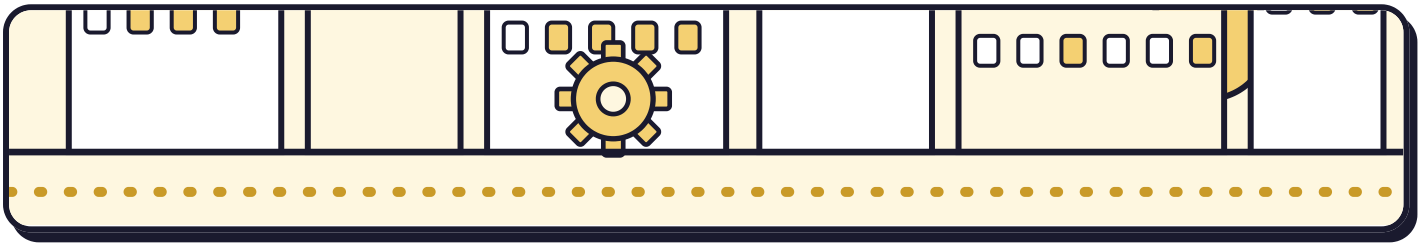
Coolest Projects is a free showcase where young people up to 18 share the things they make with technology.[1] This year's favourites show what happens when you start with someone who needs help.

**COOLEST PROJECTS 2026**

**15,000+**
young people in the online showcase[2]

**4,500+**
creations shared[2]

**40**
countries taking part[2]



The judges picked their favourite projects of 2026.[2] Here are four of them, described the way their makers described them. We have used first names only, because that is how the showcase lists them.

## Helping Leonidas breathe better

BY KATERINA AND LEONIDAS · GREECE

A home made station, built with a Raspberry Pi computer, that keeps an eye on how clean the air is indoors.[3]

**Made for a real person, and his name is in the title.**

## Sub Racers

BY HASEEB · UNITED KINGDOM

A 3D racing game made as a tribute to ocean conservation and to David Attenborough's 100th birthday.[4]

**Made to help people think about the challenges facing our oceans.[4]**

## Ornimetrics

BY BAICHEN · USA

A smart bird feeder that uses AI to recognise which kinds of birds come to visit, as they arrive.[5]

**Technology that helps you notice nature, rather than miss it.**

## HopeLink

BY DANIEL · CANADA

An app that helps people in Toronto quickly find nearby shelter, food, medical care, hygiene and transport services.[6]

**Built for the people in a city who need help the most.**

WHERE THIS CAME FROM


1 **Raspberry Pi Foundation**, Coolest Projects 2026 opens for entries in January, 14 January 2026. raspberrypi.org
2 **Raspberry Pi Foundation**, Celebrating over 15,000 young creators at the Coolest Projects 2026 online showcase, 25 June 2026. raspberrypi.org
3 **Coolest Projects**, DIY Raspberry Pi Air Quality Monitor: Helping Leonidas Breathe Better, 2026. online.coolestprojects.org
4 **Coolest Projects**, Sub Racers, 2026. online.coolestprojects.org
5 **Coolest Projects**, Ornimetrics, 2026. online.coolestprojects.org
6 **Coolest Projects**, HopeLink, 2026. online.coolestprojects.org

The Maker District YOUR TURN

# Design your own invention

Every project on page 4 began with one question: who needs help, and with what? Start with a person, not a gadget.



BLOOP SAYS

I've invented a robot that finds lost socks. It's brilliant. It's also missing a wheel. I know it's here somewhere.

ACTIVITY NO SCREEN NEEDED

## My invention blueprint

Fill in each box. Draw big, label everything, and do not worry if it would be hard to build. Every inventor starts on paper.

**01 · DRAW IT**

Draw your invention here. Add arrows and labels to show what each part does.

**02 · THE PROBLEM**

What is the problem, and who has it? Write their name or draw them.

**03 · HOW IT HELPS**

What does your invention do to make things better for them?

**04 · WHAT IT NOTICES**

Light, sound, movement, heat, or a button being pressed?

**05 · WHAT IT DOES BACK**

Does it light up, make a sound, move, or show a message?

**06 · NAME IT**

Give it a name worth remembering.

**07 · TEST IT**

How would you know it works? Who would you ask to try it?

MAKER TIPS

- Ask the person with the problem what they think. Good inventors listen first.
- Stuck? Build a model from a cereal box and some tape before you draw.
- Most gadgets work the same way: something notices, a program decides, and something happens.
- Want to make it for real one day? Page 10 has free places to start.

The Research Room

WHAT SCIENTISTS KNOW

# Not all screen time tells the same story

Grown ups often ask how long is too long. Scientists are finding a more useful question: what are you actually doing on the screen, and who are you doing it with?[1]

### Creating

Coding a game, making music, drawing a comic.

**You are the one making the choices.**

### Learning

A how to video, a maths game, a space website.

**In one big review, watching together with a parent was linked with better reading and writing skills.[1]**

### Connecting

A video call with Grandma, a message to a friend.

**Talking with people you know is different from scrolling past strangers.** See page 7.

### Watching

Shows, videos, scrolling through a feed.

**Fun, and worth noticing: did you choose it, or was it chosen for you?** Page 8.

## WHAT THE RESEARCH FOUND

In 2024, scientists brought together 102 earlier research summaries covering almost two million young people.[1] Some kinds of screen use were linked with small to moderate harms and some with benefits, so they advised weighing up each kind separately.[1]

One example: screen time was linked with weaker reading and writing skills overall, but when parents watched together with their children, the link went the other way.[1]

## WHAT IT CANNOT TELL US YET

Most of these studies spot links, not causes: two things happening together, not one making the other happen.[2] Many also rely on people estimating their own screen time.[2]

Simple labels are argued about too. A review of 40 studies, mostly of adults, found little support for 'active' social media use being good and 'passive' use being bad.[3] It tells us little about children your age.

### TALK ABOUT IT TOGETHER

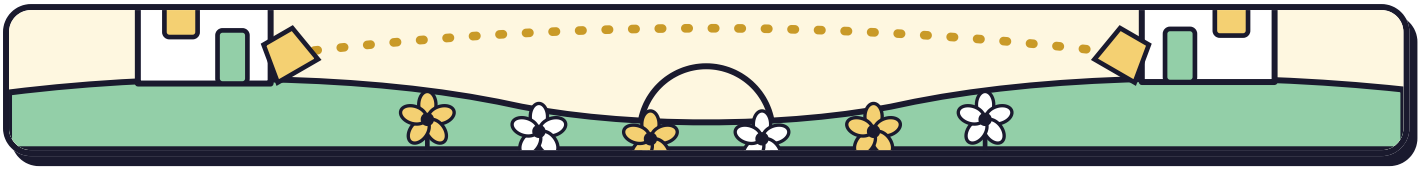
1. Think about yesterday. Which of the four kinds did you do most?
2. Which kind leaves you feeling good? Which is hard to stop?
3. What is one thing you would like to make, not watch, this week?

### DIGI SAYS

**Two things happening together does not mean one caused the other. Spotting that is a skill for life.**

### WHERE THIS CAME FROM


1. **Sanders, Noetel, Parker and colleagues, Nature Human Behaviour**, An umbrella review of the benefits and risks associated with youths' interactions with electronic screens, 2024. doi.org
2. **Kaye, Orben, Ellis, Hunter and Houghton, IJERPH**, The Conceptual and Methodological Mayhem of 'Screen Time', 2020. doi.org
3. **Valkenburg, van Driel and Beyens, New Media and Society**, The associations of active and passive social media use with well-being: A critical scoping review, 2022. doi.org



The Connection Garden · FRIENDSHIP

# Sometimes a screen helps a friendship travel further

A video call can bring a cousin from across the country into your kitchen. Scientists are learning when screens help friendships, and when they get in the way.

Lots of families use video calls and messages to stay close to people who live far away, like grandparents, cousins or a friend who has moved.

In one experiment, 130 young people aged about 12 did a stressful task. The ones who then texted a friend they knew were in a better mood afterwards than the ones who sat quietly or watched a video. [1] It was one short experiment, so it does not prove that more messaging is always better.

A UNICEF project that studied children aged 8 to 12 found that digital games can support feeling connected to other people, and that different games support different things depending on how they are designed. [2]

### KEEP YOUR GARDEN SAFE

- Connect with people you know in real life, and let a grown up know who.
- A message from someone you do not know? Tell a grown up. You are never in trouble for telling.
- Many apps, including TikTok, are for ages 13 and over. [3]



### PEBBLE SAYS

A kind message can travel a very long way. Who would smile if they heard from you today?

ACTIVITY · NO SCREEN NEEDED

## Send a friendship postcard

Cut it out, draw on the front, write on the back, then post it or hand it over.

DRAW SOMETHING YOU DID TOGETHER

DEAR .....

.....

.....

.....

.....

Add a stamp and an address with a grown up · Or take a photo of it and send it together

### WHERE THIS CAME FROM


1 **Yau, Reich and Lee, Journal of Adolescent Health**, Coping With Stress Through Texting: An Experimental Study, 2021. doi.org

2 **UNICEF Innocenti**, Responsible Innovation in Technology for Children (RITEC), phase 2 report, April 2024. unicef.org

3 **TikTok**, Terms of Service. tiktok.com

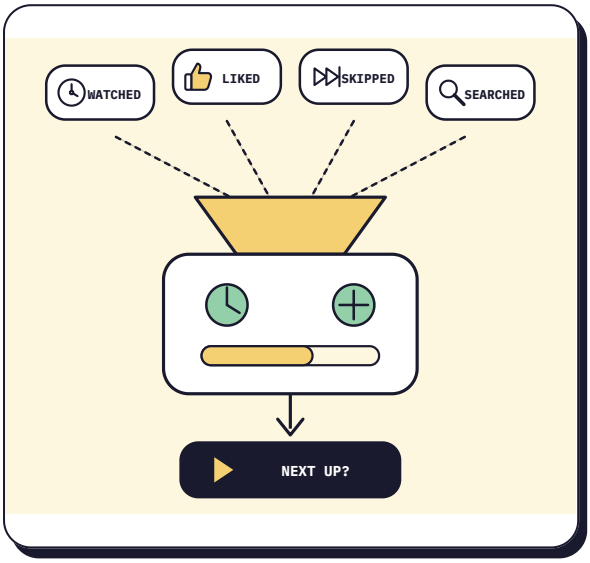
The Thinking Forest

BLOOP INVESTIGATES

# Who chose the next video? You, or the algorithm?

When one video ends and another is already waiting, somebody chose it. Bloop went looking for clues.






**BLOOP'S VERDICT**

**Case closed! The algorithm is a very fast guesser. It uses the clues you leave behind, and you are still the one holding the play button.**

So who chose? A bit of both. The algorithm makes a guess from what you did before, and you decide whether to press play.

For grown ups: a UK framework, Education for a Connected World, suggests children aged 7 to 11 learn to explain how search engines work and how their results are selected and ranked.[3]

### CLUE 1 — An algorithm is a list of rules

It is a list of steps for solving a problem or getting something done, a bit like a recipe, and the steps have to be in the right order.[1]

### CLUE 2 — It remembers what you watch

YouTube says the videos you have watched help it choose what to recommend to you.[2]

### CLUE 3 — It notices what you search for

What you search for changes what it suggests later on, too.[2]

### CLUE 4 — You can tell it no

Pressing 'Not interested' helps it learn what to stop suggesting.[2]

**TRY THIS, NO ACCOUNT NEEDED**

- Next time you watch with a grown up, look at the suggestions and guess why each one was picked.
- Play Become the Algorithm on the next page, with paper cards.
- Ask what it never shows you. Things you have not looked for yet are hiding there.

**WHERE THIS CAME FROM**

1 **BBC Bitesize**, What is an algorithm?. bbc.co.uk
2 **YouTube Help**, How YouTube recommendations work. support.google.com
3 **UK Council for Internet Safety**, Education for a Connected World, 2020 edition, 2020. assets.publishing.service.gov.uk

The Thinking Forest · A PAPER GAME

# Become the Algorithm

One of you is the Watcher and one of you is the Algorithm. Can the Algorithm work out what the Watcher likes, using nothing but clues?



**BLOOP SAYS**

I played the Algorithm and kept showing football. My friend wanted dinosaurs. I needed more clues!

**PLAYERS · 2 TO 6**

**You need:** the 12 video cards, a pencil, the notebook below.

## Set up

1. Cut out the 12 video cards and shuffle them.
2. The Watcher secretly picks a favourite topic and tells nobody.
3. The Algorithm deals three cards face up. These are the suggestions.

## Play five rounds

1. The Watcher picks one card to watch, or skips all three.
2. The Algorithm writes down what was watched or skipped.
3. The Algorithm uses the clues to choose the next three cards.
4. After round five, the Algorithm guesses the favourite topic. Swap jobs and play again.

**THEN TALK ABOUT IT**

- How quickly did the Algorithm work out the favourite?
- Which cards were never shown? Why not?
- If the Watcher skipped everything, what could the Algorithm learn?
- Real apps use clues like these, only far more of them and far faster.

**CUT OUT THE 12 VIDEO CARDS**

| | | |
|---|---|---|
| **Puppy tricks** ANIMALS | **Dinosaur facts** ANIMALS | **Best goals** FOOTBALL |
| **Keepy uppy** FOOTBALL | **Rocket launch** SPACE | **Planet tour** SPACE |
| **Draw a dragon** ART | **Clay models** ART | **Drum along** MUSIC |
| **Song quiz** MUSIC | **Cupcake bake** COOKING | **Pizza faces** COOKING |

**THE ALGORITHM'S NOTEBOOK**

| | ROUND 1 | ROUND 2 | ROUND 3 | ROUND 4 | ROUND 5 |
|---|---|---|---|---|---|
| WATCHED | | | | | |
| SKIPPED | | | | | |
| MY GUESS | | | | | |

**GOING FURTHER**

- Older readers can try Common Sense Education's free lesson AI Algorithms: How Well Do They Know You?, made for ages 11 to 18. [1]
- Make your own cards with your family's favourite topics, and play again.

**WHERE THIS CAME FROM**

[1] Common Sense Education, AI Algorithms: How Well Do They Know You?. commonsense.org

The Family Explorer FIVE GOOD THINGS

# Five good things to explore together

All free and all checked by us for this issue. Services change, so look again together before signing up, and use a grown up's account wherever one is needed.



ORBIT SAYS

I tried number 4 on a clear night. Wrap up warm, and give your eyes time to get used to the dark.

## 1 micro:bit MakeCode

Free, open source block coding for the BBC micro:bit, with a simulator that shows what your program will do. [1]

**New this year: screen reader support for block coding, shaped with blind and low vision young people.** [2]

WHO FOR Beginners, with a grown up
ACCOUNT Ask a grown up
makecode.microbit.org

## 2 Scratch

Snap code blocks together to make games, stories and animations. Free for everyone. [4]

**Start small: make a cat move, then make it dance.**

WHO FOR Ages 8 to 16 [3]
ACCOUNT Ask a grown up first
scratch.mit.edu

## 3 ESA Kids

Space news, facts, animations and downloads from the European Space Agency. [5]

**Liked page 3? Find out more about life in space.**

WHO FOR Ages 6 to 12 [5]
ACCOUNT None
www.esa.int/kids/en/home

## 4 Night sky highlights

Free guides from Royal Museums Greenwich to what is up in the sky. [6]

**Looking up needs no screen at all. Read it together, then go outside.**

WHO FOR Everyone
ACCOUNT None
OFFLINE Jot the highlights on paper
www.rmg.co.uk/stories/space-astronomy

## 5 Code Club projects

Free step by step coding and making guides from the Raspberry Pi Foundation. [7] Code Clubs are free for all school aged young people. [8]

**Ask your school or library about a club near you.**

WHO FOR School aged young people
ACCOUNT Not to read the guides
projects.raspberrypi.org

WHERE THIS CAME FROM


1 **Micro:bit Educational Foundation**, MakeCode. tech.microbit.org
2 **Micro:bit Educational Foundation**, MakeCode screen reader support, 13 July 2026. microbit.org
3 **Scratch Foundation**, Scratch, frequently asked questions. scratch.mit.edu
4 **Scratch Foundation**, About Scratch. scratch.mit.edu
5 **European Space Agency**, ESA Kids. esa.int
6 **Royal Museums Greenwich**, Night sky highlights, October 2026, 29 September 2026. rmg.co.uk
7 **Raspberry Pi Foundation**, Learn. raspberrypi.org
8 **Code Club**, Code Club. codeclub.org

The Family Explorer

THE FAMILY EXPERIMENT

# Seven days of curiosity

One small thing a day for a week. Tick a circle for everyone who joins in, then compare notes on day seven.



BLOOP SAYS

Thursday is my favourite. I'm still looking for the last piece of mine.

### MON

**Make a pixel picture**

Colour the 8 by 8 grid on page 3. Can a grown up guess what it is?

OFFLINE

### TUE

**Five minutes of sky**

Go outside after dark with a grown up and count the stars you can see.

OFFLINE

### WED

**Spot the guess**

Watch one thing together. Talk about why the next suggestion appeared.

TOGETHER

### THU

**Fix a problem**

Ask someone at home what annoys them, then design a fix on page 5.

OFFLINE

### FRI

**Send a smile**

Finish your friendship postcard from page 7 and send it.

OFFLINE

### SAT

**Learn it together**

Find a how to video, watch it together, then try it for real.

TOGETHER

### SUN

**Show and tell**

Everyone shows one thing they made or found out this week.

OFFLINE

**THE BEST THING WE MADE**

**SOMETHING THAT SURPRISED US**

**WHAT WE WILL KEEP DOING**

The Family Explorer

CUT OUT AND KEEP

# Your Planet Passport

Collect a stamp for every activity you try in this issue. Colour in each circle once you have done it, then ask a grown up to sign it off.



PLANET PASSPORT

## Issue 001, The Curiosity Edition

This passport belongs to an explorer of The Guided Planet.
Stamps are collected by doing things, never by buying them.

EXPLORER

FIRST NAME ONLY

- Pixel picture — PAGE 3
- Inventor — PAGE 5
- Postcard sent — PAGE 7
- Algorithm — PAGE 9
- Explorer — PAGE 10
- Seven days — PAGE 11

SIGNED OFF BY A GROWN UP

DATE

NEXT MONTH · DECEMBER 2026

## The Real or Pretend Edition

- How to tell when a picture might have been made by a computer
- Orbit's guide to checking a story before you share it
- A winter sky to look out for
- A new stamp for your Planet Passport

A LITTLE MORE WONDER. A LOT MORE UNDERSTANDING.

EVERY SOURCE IN THIS ISSUE

Each page lists where its facts came from. The full list, with links, is online.

www.guidedchildhood.com/guided-planet/issue-001/sources

FOR GROWN UPS

The Guided Planet is made by The Guided Digital Childhood. We help families get children to sixteen ready for a phone and social media, ten minutes a day, together.

**What stage is your child? Three questions, no sign up.**

www.guidedchildhood.com/starter-pack

The Guided Planet

The Guided Planet is made by The Guided Digital Childhood and is free to print, copy and share with your class or club. Our philosophy: delay where it helps, protect where we can, prepare throughout.

ISSUE 001
NOVEMBER 2026
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